

The stories, science, and data behind the artwork



04

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AI

latest research

What you see. What it means. Why it matters.

Contents / A shared reference

Facts, meanings, sources, country stories, and optional ways to frame the ideas.

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About this guide

Artist-team reference | For curious eight-year-olds | English Last updated: 2026-09-09.
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Facts, comparisons, and country stories for curious eight-year-olds. Artists can explain these ideas in their own words. Examples are optional; **artist notes** preserve source details.

An artwork about people and the world they live in

A sore back can make walking difficult. Worry can make concentrating difficult. Changes around a home affect people and nature. The globe connects these concerns: caring for people also involves caring about their surroundings.

P.A.I.N. means **Personal And Interconnected with Nature**. Its longer map name is the Personal-Planetary-Pain Map. Artist Mary Maggic leads the project with collaborators in art, sound, design, technology, data research, and network medicine.

Four layers, four questions

The layers work like pictures on clear sheets. Each adds information, while showing a different part of the story.

Layer	What it helps us explore
Physical	How painful body conditions can affect everyday life.
Emotional	What kinds of distress appear in selected writing.
Environmental	How temperatures change and activities produce emissions.
Socioeconomic	One way to compare economic resources around people's lives.

The artwork makes **data**, recorded information, visible through shapes, colours, and words. The marked surface suggests injury, although Earth does not feel pain as a person does. There is no single score deciding which country suffers most.

Data, estimates, and samples

A **measurement** records something observed, such as a temperature. An **estimate** uses available evidence to work out a value when every detail cannot be measured directly.

A **sample** is a selection from something larger. Reading one shelf tells us about those books; another shelf may contain different stories. The emotional layer also uses a selection.

Artist note: Country symbols show available data. Physical and environmental symbols use the strongest local value, not a national average. Brightness can indicate selection. Small places may have words but no selectable area on the simplified map.

Learning to read the picture

An ordinary map might show roads with lines. This artwork has its own visual language, connecting what we see with what it represents.



Callouts: 1, emotional words; 2, physical surface; 3, environmental atmosphere; 4, economic yellow; 5, layer choices. Exhibition image: 9 September 2026.

What appears	What it represents
Dents and dots	Information about painful health conditions. A dot is not one patient.
Red atmosphere	Temperature change compared with an earlier climate.
Pale or green atmosphere	Emissions associated with activities, given an artistic shape.
Yellow countries	Lower economic output per person gives stronger yellow.
Words	Themes the AI found in writing associated with a country.
Curving lines	Geographic connections among places sharing a displayed theme.

Movement and sound

Rotation, zoom, and country selection bring information into focus. Emotional filters change which themes can appear using the same texts. Moving lines help our eyes follow connections.

Composed music and interaction tones direct attention; they are not recordings of pain. Projection mode hides the separate web survey. Participation requires no personal disclosure.

Physical pain / Why population and place matter

Reading, playing, sleeping, and meeting friends all depend partly on how our bodies feel. Pain can interrupt these activities even when other people cannot see it. The physical layer makes some of that hidden difficulty visible.

Six health categories

Condition	What the name means
Headache disorders	Conditions involving headaches that can return again and again.
Migraine	A condition involving the nervous system, often with headache and other symptoms.
Low back pain	Pain around the lower back that can make movement difficult.
Neck pain	Pain that can affect turning the head and moving comfortably.
Osteoarthritis	A condition affecting joints, where bones meet, causing pain or stiffness.
Rheumatoid arthritis	An illness involving the body's immune system that can inflame joints.

The immune system usually defends the body against illness; rheumatoid arthritis involves it acting in a harmful way. Migraine belongs within the wider headache category, so these categories do not count separate groups of people. [C3, C5-C7]

A made-up example: more people, the same fraction

Imagine one town with **100 people** and another with **1,000 people**. In each town, **one person in every ten** has the same painful condition.

Imaginary town	People affected	Fraction affected
100 residents	10 people	1 in 10
1,000 residents	100 people	1 in 10

The larger town has ten times as many affected people. It may need more care services even though the fraction affected is unchanged.

Population information helps place the health data, contributing to strong marks near dense parts of India. The town numbers are invented examples, not measurements from India.

Artist note: IHME Global Burden of Disease and WorldPop are the source families. The selected layer contains **15,282 condition-location observations, 2,547 per category**; 13,183 are assigned to 155 countries. These are neither interviews nor patient counts. The original release, reference year, health measure, upstream size, and full transformation remain unverified. WorldPop's described family covers 2000-2020 at roughly 1 km resolution, but a selected year is not confirmed. [S7-S8]

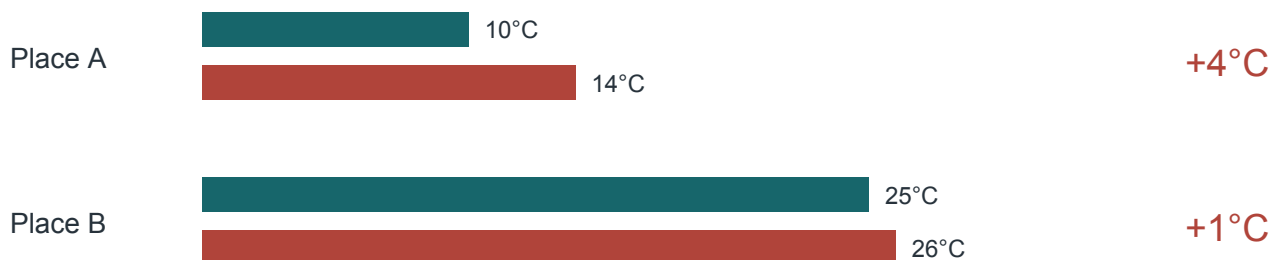
Environment / Warming, heat, and emissions

Weather is what happens outside over a short time, such as a rainy afternoon. **Climate** describes patterns over many years. To investigate warming, researchers compare temperatures across periods. A warm afternoon by itself cannot tell the whole climate story.

A made-up example: which place changed more?

Imagine two places with these average temperatures in an earlier and a later climate period. An average combines several values to give one summary; these are not temperatures on one day.

Which place is warmer? Which warmed more?



Teal: earlier climate average. Coral: later climate average.

Made-up numbers for learning; these are not exhibition measurements.

The warmer place is still warmer. The cooler place has warmed more: 14 minus 10 is 4, while 26 minus 25 is 1. That is why a cold country can have a strong warming signal. These numbers are invented examples, not measurements for Russia or Canada.

What emissions mean

Emissions are substances released by activities, such as gases released when fuels burn. Producing energy, making goods, and transport can produce emissions. The exhibition uses a floating form to make this environmental pressure visible. It is not a measured cloud of smoke.

Warming asks how a place has changed; emissions concern what activities release. Country borders locate information but do not form walls around the air. The atmosphere gives a shared concern a shared shape.

Artist note: Berkeley Earth supplies **8,100 selected temperature observations**; 2,591 are assigned to 140 countries. Climate TRACE supplies **2,339 emissions observations**; 1,873 are assigned to 146 countries. Original dataset sizes, selected releases, temperature baseline and comparison period, and precise emissions measure remain unconfirmed. Neither colour nor height has a verified conversion to degrees, tonnes, or personal exposure. [S9-S10]

Economic circumstances / What an average can tell us

An economy includes making goods and providing services. Food production and building a school are examples. **GDP** adds up the value produced within an economy. **GDP per person** divides that total by the number of people living there. It gives one comparison of economic output.

The globe uses stronger yellow for lower GDP per person. Resources can help make schools, transport, and care possible, but the average alone does not tell us who can use them.

A made-up example: total and per person

Imagine two groups producing things worth units in an imaginary economy.

Group	Total value made	Number of people	Value per person
Smaller group	12 units	3	12 divided by 3 = 4
Larger group	20 units	10	20 divided by 10 = 2

The larger group makes more altogether; the smaller group makes more per person. These answer different questions.

The smaller group's 12 units might belong mostly to one person. Dividing by three does not make everyone receive four. GDP per person is not money handed to each resident.

What the yellow leaves open

Resources, prices, and public services differ between people and places. Production cannot measure kindness, happiness, dignity, or friendship. Other layers offer other questions.

Artist note: World Bank 2024 GDP per capita, current US dollars, supplies 190 values and 16 missing entries. The display compresses the wide range using a logarithmic scale. That does not correct for unequal sharing or differences in purchasing power. [S11]

Emotional words / Patterns in writing

A sentence can describe loneliness and fear together. The emotional layer looks for themes in news, posts, and complaints. A **category** groups related ideas. The AI can recognize several categories in a passage even though the globe gives one word prominence.

Fourteen broad categories

Category	Meaning in this project
Body pain	Writing expressing physical hurt.
Hurt	Emotional pain without a more particular description.
Eco-anxiety	Worry about environmental change.
Uncertainty	Distress about an unstable situation or an unclear future.
Grief	Distress over loss, including changes to a familiar home environment.
Anger	Anger or outrage, sometimes connected with something felt to be wrong.
Hardship	Difficulty getting what is needed, or coping with insecurity and loss.
Displacement	Distress connected with being uprooted from a familiar home.
Trauma	Distress described after overwhelming experiences.
Loneliness	Pain around connection, relationships, and feeling alone.
Depression	Language describing depressive distress, beyond an ordinary brief disappointment.
Fear	Fear, anxiety, or panic.
Helplessness	Feeling unable to influence what is happening.
Shame	Shame, guilt, or blaming oneself.

These categories describe writing, not national feelings or diagnoses. Two stories can share Anger while describing different situations.

How a country gets a word

For news, the country is where the **publisher** is based. A publisher in India can report on an event in another country. Other sources use supplied country or location information. The map therefore collects writing associated with a place, not interviews with all residents.

Combined text scores summarize the selected writing. The strongest of the fourteen distress categories supplies the word: a pattern in texts, not a percentage of people.

A language choice is also an artistic choice

Artists choose each word's language: Hindi for India, Kirundi for Burundi, Palauan for Palau. Articles may use other languages. One choice cannot represent every language community.

Artist note: The current display uses **249,438 texts: 5,000 news articles and 244,438 other texts**, covering 192 countries. Where both groups exist, the news average gets 75 percent of the weight and the other-text average 25 percent. A sole available group gets full weight. Anger leads in 152 countries. There are 84 source-language hints and 101 displayed language codes, distinct from the classifier's 100-language coverage.

Artist note: No-pain and out-of-scope scores do not compete for the displayed word. At least one of those controls exceeds the visible distress winner in 121 of 192 countries. A prominent word can therefore accompany weak distress evidence. Filters can change the winning eligible category without changing the texts.

Connections / Why the globe has lines

A **network** is a set of points joined by lines. Here, each point represents a country label. Countries with the same displayed emotional category belong to the same group. Geography helps decide which pairs get a direct line, so the artwork does not need every point to join every other point.

Delaunay: a mesh of neighbours

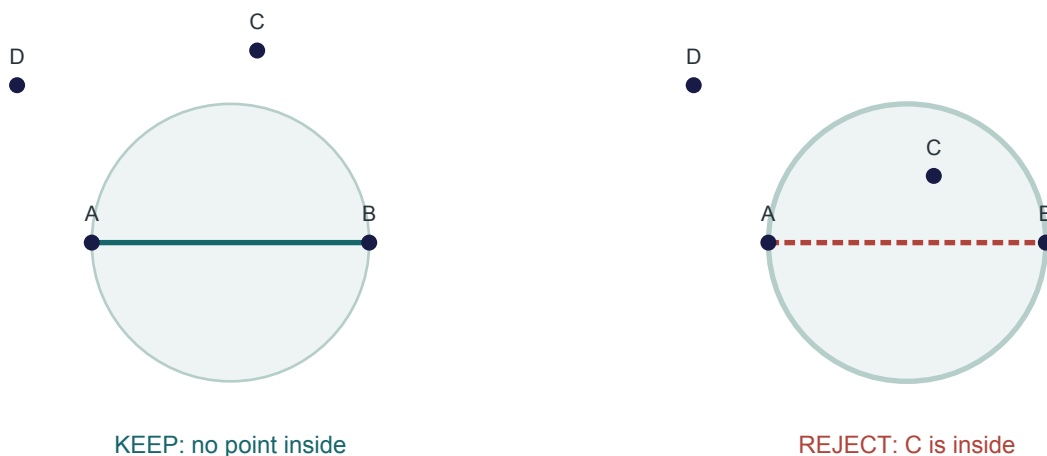
The exhibition currently uses a **Delaunay** network. It arranges places sharing a category into a mesh of triangles around the globe. Triangle corners are country-label positions; their sides supply connections. The lines follow curves around the globe's surface.

Some places connect through other points rather than a direct line. This arranges a shared theme geographically; it does not show friendship, trade, or one country causing distress.

Gabriel: leaving out some connections

A related mathematical idea is called a **Gabriel graph**. On a flat drawing, imagine a circle with two matching-category points at opposite ends of its diameter. If another point from that same category lies inside the circle, the pair does not get a direct Gabriel connection. A point from a different category does not block it.

The idea behind a Gabriel connection



A flat illustration of proximity; the globe uses the same idea in three dimensions.

The globe uses a corresponding three-dimensional ball. This check usually leaves fewer lines, changing connections while keeping the same category scores and writing.

Artist note: Current complete data has 192 nodes and 522 Delaunay edges across all categories; the Gabriel comparison has 328. A selected view need not show all edges together. For groups of fewer than four nodes, the exhibition comparison joins all pairs, so it does not perform the full mathematical Gabriel check there. The diagram explains the mathematical idea. Animated connections direct attention; they are not a history of spreading emotion.

PCAI / How the AI learns to classify text

PCAI prepares country-associated writing and uses an AI **classifier** to identify distress categories. It can examine far more writing than one visitor could read. People still need to investigate what the results mean.

Teaching examples in 100 languages

The classifier is designed to classify writing in **100 languages**. Its **synthetic dataset is multilingual**: constructed teaching examples use many languages, rather than English alone. Synthetic means made for this teaching purpose. A fictional sentence about missing a friend is not testimony from a real child.

Teaching links wording and context with categories. The trained model then examines real texts, called classification or inference. Reading more articles does not retrain it.

Words need context

Consider these invented sentences: "I am afraid" and "I am not afraid." One small word changes the meaning. "The character said, 'I am afraid'" describes a character's words; it does not necessarily describe the writer. "I miss my friend and worry about tomorrow" might fit both loneliness and uncertainty.

The model looks for patterns beyond a list of special words. Negation, quoted speech, unusual phrasing, and several themes together can still confuse it. Languages can express similar ideas in different ways. Being designed for 100 languages does not mean the same reliability in each one. The complete language list appears in the artist appendix.

PCAI / Teaching and testing are different jobs

Researchers test with separate examples, like practising maths problems and then checking learning on different problems. Harder tests reveal difficulties that easy examples miss.

Stage	Examples	Meaning
Prepared synthetic collection	573,143	Multilingual teaching resource covering 100 languages.
Used to fit the adopted model	333,236	Examples actually used to teach its smaller network.
Clean synthetic test	15,874	Separate evaluation examples.
Harder synthetic stress test	10,429	Challenging withheld evaluation examples.

Artist note: Adopted **version 9**, finalized **4 September 2026**, uses pretrained **BGE-M3** to represent language numerically, followed by a project-trained network with 1,024 hidden units. Its 70 related decisions include 3 relevance, 14 broad-category, 47 fine-category, 2 safety, and 4 out-of-scope-reason decisions. These are not 70 emotions. It reads at most 512 tokens, pieces of text that are not always whole words; later article context can be missed. The fitted examples comprise 295,572 newer and 37,664 earlier reviewed examples. The table's stages are not counts to add together. [\[S12\]](#)

Artist note: Broad/fine macro-F1 is **0.9028/0.7772** on the clean synthetic test and **0.6658/0.6552** on the stress test. F1 balances finding labelled examples with avoiding false alarms; macro gives categories equal weight. These results are not percentages of people understood correctly, nor verified accuracy across every real-world language.

CCNews / From a huge library to a research collection

CCNews contains international news gathered through Common Crawl and prepared by Stanford OVAL. The broader source is described as **about 600 million articles** in more than 100 languages. PCAI's country-linked main collection has **about 394 million**. That is the resource often described as roughly 400 million articles. [S1]

These numbers describe different shelves in a huge library. PCAI acquired a snapshot and prepared country-linked texts. The upstream approximation is not an exact download count.

What preparation changes

Duplicates can make one text seem more common. Researchers reduce repeated copies, standardize text, check dates, and mask detected identifiers such as emails and phones. Unclear or conflicting publisher-country assignments stay out of the country-linked collection.

This is **general news**, with no pain word required. The classifier later assesses distress. Publisher output, language coverage, and country identification affect which voices remain.

Artist background: exact collection stages

Stage	Articles or records	What it counts
Broader Stanford OVAL source	About 600 million	Approximate upstream collection.
Locally acquired snapshot	474,985,436	Records actually acquired and counted.
Prepared texts	472,438,058	After duplicate, date, and privacy preparation.
Publisher country resolved	394,231,021	Usable, unconflicted country associations.
Main corpus	393,745,301	Final country-linked collection; 186 countries.
Its recorded years 2015-2026	388,842,327	Included within the main corpus.

Artist note: The remaining **4,902,974** main-corpus records are dated 2010-2014. Acquired and prepared texts carry **116 language codes**. Recorded dates may be publication, crawl, or substituted acquisition metadata. The main corpus has no 2025 entries; a 2026 date alone does not prove fresh reporting. The upstream crawl description covers 2016 to June 2024.

Filtering stage	Records excluded	Purpose
Initial duplicate checks	906,602	Reduce repeated text.
Initial date checks	1,640,769	Apply supported date rules, not a 2015-2026-only filter.

Filtering stage	Records excluded	Purpose
Later privacy cleaning	7	Remove additional duplicates revealed by masking.
Publisher-country checks	78,207,037	Exclude unresolved or conflicting country attribution.
Prose, privacy, and eligibility	485,075	Retain usable text and consistent records.
Final duplication/conflict checks	645	Avoid counting repeated or conflicting material again.

Artist note: Most masking changes text rather than removing an article. It covers detected emails, phones, links, and handles, without guaranteeing complete anonymity. Different retained articles can still discuss the same event. The final main corpus has no per-country cap. Preparing it and completing classification are separate achievements.

Sources / What is on display and what is newer

The exhibition and the latest research use different selections. The globe's current emotional words come from **249,438 texts**. A newer completed analysis examines **26,176,468 texts**. The larger research result has not replaced the displayed selection.

What the six text sources contribute

News reflects publisher coverage. Social posts offer other expression. Civic tweets concern public decisions; consumer complaints describe financial-service problems. Different purposes can produce different themes even within one country. [\[S1-S6\]](#)

Source	Exhibition records	Latest completed records	Countries
Stanford OVAL CCNews	5,000	25,932,030	186
Twitter Sentiment by Country	166,837	166,837	163
DEMOTEC civic-participation tweets	67,193	67,193	136
Yachay coordinate-linked texts	8,564	8,564	4
CFPB consumer complaints	1,030	1,030	1
CMU GeoText microblogs	814	814	1
Combined	249,438	26,176,468	192

Country coverage overlaps. Many texts can come from the same person or discuss the same event. These totals count records, not people interviewed.

Artist background: dates and language metadata

Source	Recorded years	Language hints
Earlier exhibition CCNews selection	2015-2021; 2023	59
Latest completed CCNews selection	Mostly 2017-2024	104
Twitter Sentiment by Country	2020	60
DEMOTEC	2015-2021	9
Yachay	2014-2022	47
CFPB	2025	2
CMU GeoText	2010	8

Artist note: Hints describe likely source language, not guaranteed accuracy. Yachay's broader source reports over 600,000 posts across six countries; its selected 8,564 records cover four. Original selected-release totals for Twitter Sentiment, DEMOTEC, CFPB, and CMU remain unverified. The five smaller sources combine to 244,438 records across 175 countries.

The latest completed classification

The completed news run analysed **25,932,030 articles** from the main corpus, aiming for up to 250,000 per country. Those articles are already included in the 393.7-million main collection. Adding the five smaller sources produces the **26,176,468-record** research result.

With each record weighted equally, news supplies **99.07 percent** globally. Anger or moral injury leads in **178 of 192** country results, versus 152 in the exhibition. Selection and weighting change patterns; more writing alone does not make a population survey.

Artist note: India's latest result contains **251,442 records**, including 250,000 news articles. **94.2 percent** of the latest news records are dated 2017-2024. The verified completed result was summarized by 8 September 2026. As checked on 9 September, available records do not establish that production classification of all 393.7 million articles has started. The broader programme is ongoing; no full-corpus progress percentage is claimed.

Country stories / Twenty-five ways to explore

These stories describe the **current exhibition**. Strongest and smallest concern the named layer among places with available data, never all suffering. Physical and environmental comparisons use the strongest local signal. Notes give observation counts, not patient totals.

Story 01 / India: many people, many daily interruptions

India has a strong migraine-related signal. Where many people live close together, many lives can be interrupted by pain. The imaginary towns explain how this can happen without a higher fraction affected. A child studying and someone caring for a relative may need different help with similar pain.

Evidence: 266 physical observations, including 45 migraine observations. Population helps place the health data; a high individual risk is not established.

Story 02 / Pakistan: when a headache keeps returning

Pakistan has the strongest selected headache and migraine signals. A returning headache can interrupt a favourite book repeatedly, helping explain a familiar condition's large effect on daily life. Migraine belongs within the wider headache subject, so adding the two categories as different people would count some twice.

Evidence: 13 observations in each category; highest among 149 headache and 148 migraine country comparisons. National causes are not established. [C7]

Story 03 / Japan: movement and independence

Japan has a strong back-pain signal. Getting out of a chair and walking to meet someone help people stay independent. Almost three in ten residents were 65 or older in 2024. Age matters in back-pain research, although its exact contribution to this particular mark is not known.

Evidence: 17 back-pain observations; strongest of 150 country signals. Population aged 65 or older: 29.3 percent in 2024. Age is contextual evidence. [C3, C8]

Story 04 / China: one joint, different conditions

China has strong signals for two forms of arthritis. Joints include knees and fingers. Osteoarthritis and rheumatoid arthritis affect them through different processes, but both can make holding a cup or climbing stairs harder. The shared word suggests a connection; the different names help distinguish the conditions.

Evidence: 135 osteoarthritis and 137 rheumatoid-arthritis observations. Each has the strongest signal in its selected country comparison, with 149 and 153 countries. [C5-C6]

Story 05 / Egypt: turning towards someone

Egypt has a strong neck-pain signal. Turning towards someone who called our name seems simple when it feels comfortable. Pain can make that turn difficult, affecting reading or travel too. The example connects a distant mark with familiar movement without identifying a job or habit that caused Egypt's result.

***Evidence:** 12 neck-pain observations; strongest among 148 country signals. No specific local occupational or lifestyle cause has been established.*

Story 06 / United Kingdom: what appearances leave out

The United Kingdom has a strong rheumatoid-arthritis signal. Someone may look well while their joints feel painful or swollen. Opening a jar or fastening clothing can take extra effort. What we see from outside does not always match what a body experiences, which helps explain why support matters.

***Evidence:** 9 rheumatoid-arthritis observations; second-highest among 153 country signals. The condition involves immune-related inflammation, not a diagnosis made by the map. [C6]*

Story 07 / Guyana and Vanuatu: quieter does not mean complete

Guyana and Vanuatu have low physical signals. This layer covers six conditions, not every kind of pain. A school chart recording reading books cannot tell us about music or sport. Similarly, a quiet physical mark answers one question while leaving experiences outside the selection.

***Evidence:** 12 physical observations per country, two per condition. Their overall local maxima are second- and third-lowest among 155 observed countries.*

Story 08 / Russia: cold and changing

Russia has the strongest local warming signal here. Cold places can warm substantially, just as the imaginary cooler place warmed more. Arctic research describes interacting processes, including snow and ice changes. Russia contains many climates, so this regional context does not explain every place within its huge area.

***Evidence:** 363 temperature observations; highest local signal among 140 countries. Arctic research supplies regional context, not the selected observation baseline. [C1]*

Story 09 / Canada: beyond today's coat

Canada also has a strong warming signal. Wearing a coat tells us about today's conditions; the warming layer concerns change across years. A familiar landscape can have a changing climate. This connects an everyday decision, what to wear, with a longer question about surroundings that people, plants, and animals depend on.

***Evidence:** 220 temperature observations; second-highest local signal among 140 countries. The result is not a uniform temperature increase across Canada. [C1]*

Story 10 / Uruguay: a zero with a precise meaning

Uruguay's temperature signal is zero in this picture, with observations present. That differs from an empty entry. It does not mean Uruguay has never warmed. We still need to confirm which earlier period the source comparison uses. The zero answers a particular question; it cannot tell us about every environmental change.

***Evidence:** Two temperature observations; zero local maximum, tied at the observed minimum. It is a transformed display value, not zero degrees of climate change.*

Story 11 / Iceland: a place is more than its name

Iceland has a very small temperature signal here. Its name brings ice to mind, but a name is not a measurement. Weather, landscape, warming, and how easily a place could be harmed describe different things. A faint mark invites investigation of this comparison rather than a decision about everything changing in Iceland.

***Evidence:** Three temperature observations; smallest positive local maximum among 140 observed countries. The comparison does not summarize Iceland's entire climate.*

Story 12 / Russia: a different environmental question

Russia also has a strong emissions signal. Emissions concern substances released by activities; warming concerns temperature change. Both layers can describe the same country without measuring the same thing. The floating atmosphere connects local activities with shared surroundings: air does not stop at a line on a political map.

***Evidence:** 273 emissions observations; strongest local signal among 146 countries. It is neither a national emissions total nor a measure of what someone breathes. [S10]*

Story 13 / Iran and Iraq: activity made visible

Iran and Iraq have strong emissions signals. Making electricity, transporting goods, and producing materials can release substances. The artwork makes that pressure noticeable. It does not establish which activity explains these marks; that needs specific evidence about the selected observations.

***Evidence:** Iran has 22 emissions observations and Iraq five; their local maxima are third- and fifth-highest among 146 countries. No sector is identified by the visual. [S10]*

Story 14 / Guinea-Bissau: low emissions, another open question

Guinea-Bissau is near the low end of the emissions view. What a place releases differs from the environmental changes it experiences. A low mark cannot establish that people or ecosystems face little environmental danger. One observation also leaves many local details outside this particular picture.

***Evidence:** One emissions observation, near the low end of 146 country signals. There is no assigned temperature observation or selected emotional observation for this country.*

Story 15 / Burundi: resources around a life

Burundi has the strongest yellow because its GDP per person is lowest here. Resources can help provide schools, transport, and care. World Bank context describes development constraints, including low-productivity agriculture. This explains some economic background, while the colour leaves each family's resources and access to help unknown.

Evidence: One 2024 GDP value, about US\$216 per person; lowest of 190 available values. Country context is not a calculation of each cause's contribution. [C2, S11]

Story 16 / Luxembourg: who contributes to the total?

Luxembourg has high GDP per person and weak yellow. Cross-border workers contribute to its production while living outside the resident population used in the division. An average therefore needs both parts understood: what goes into the total, and which people are counted underneath it.

Evidence: One 2024 GDP value, about US\$137,782 per person; third-highest of 190 values. Cross-border worker accounting is verified context, not a typical resident's salary. [C4]

Story 17 / Monaco: resources and feelings

Monaco has the highest GDP per person here and the weakest yellow. It also displays Hurt. Production and emotional language answer different questions. A fictional person can have resources and still miss a friend. This explains how the layers differ without claiming what Monaco's residents actually feel.

Evidence: GDP about US\$288,002 per person, highest of 190 values. Hurt comes from 27 news articles and 1,034 other texts; these are separate kinds of evidence.

Story 18 / India: a large total and a large population

India separates total economic size from output per person. The imaginary larger group made more altogether but less per person. A large country can likewise have a large total and many residents. Yellow uses the per-person comparison, leaving another question open: how are resources shared within the country?

Evidence: One 2024 GDP value, about US\$2,592 per person, forty-sixth-lowest of 190 values. This is not total GDP or the percentage of people experiencing poverty. [S11]

Story 19 / Yemen: a missing piece

Yemen has no available GDP value here, although Grief is present in its emotional layer. Each source brings different information. A missing entry means this evidence cannot answer that question, like a question left unanswered on a worksheet rather than an answer of zero.

Evidence: No selected 2024 GDP observation; the emotional observation remains present. Missing economic data does not mean no economic hardship.

Story 20 / Burundi: anger at something unfair

Burundi's texts have a strong Anger result, a category including outrage at perceived wrongdoing. A fictional person might be angry because a friend was treated unfairly. This helps explain the category without calling a population aggressive. The actual texts' reasons require evidence beyond their label.

***Evidence:** 28 news articles plus two other texts; highest selected anger score among 192 countries. The exhibition weights the two source-group averages 75/25.*

Story 21 / Palau: one article is a small window

Palau's Uncertainty word comes from one article. One page can tell an interesting story without explaining a country's life. The AI gave a higher score to "out-of-scope," meaning writing that does not fit the question being asked. The map still chooses among distress words, so a visible word need not mean strong evidence of distress.

***Evidence:** One publisher-linked news article, no other texts. Uncertainty score 0.1557; out-of-scope score 0.6492. Neither value is a percentage of residents.*

Story 22 / Timor-Leste: one category, several possible stories

Timor-Leste has a strong Anger theme in eight texts. An unfair rule, a disagreement, or a loss could produce angry language in fictional stories. The category does not tell us which explanation fits the actual writing. Eight texts give a starting point while leaving many experiences unrepresented.

***Evidence:** Five news articles plus three other texts; second-highest selected anger score among 192 countries. National causes are not established.*

Story 23 / Sao Tome and Principe: concern about a changing home

Sao Tome and Principe shows Eco-anxiety. A fictional character might worry about changes to a familiar environment. Such language expresses concern; a temperature record supplies a different kind of evidence. Together the layers distinguish changes around a place from how concerns appear in writing.

***Evidence:** 27 publisher-linked news articles, no other-text contribution; highest eco-anxiety score among 192 countries. It is not a population anxiety rate.*

Story 24 / Oman: pain in words and health data

Oman's emotional word is Body pain, found in writing. Its physical surface comes from health estimates. A sentence about a character's sore back and a study of back-pain burden can both matter while answering different questions. Their shared subject does not make the text score a patient count.

***Evidence:** 27 news articles plus 968 other texts; highest body-pain language score among 192 countries. The separate physical layer has 36 mapped observations.*

Story 25 / Comoros: visible through one source

Comoros has economic information but no selected emotional texts. Imagine a library with a geography book about a place but no storybook from there. The missing book limits the collection; it does not mean the place has no stories. Sources determine what the globe can make visible.

Evidence: GDP about US\$1,858 per person in 2024; zero news and zero other-text records in the emotional selection. Missing emotional evidence is not evidence of no distress.

Useful words and optional connections

Maps, stories, maths, and art offer different ways into the same facts. Comparisons are optional, with no required route.

Possible connections include transparent sheets for layers, library shelves for samples, and fictional characters for overlapping feelings. The worked examples explain numerical comparisons. Each leaves room for a visitor's own questions.

A working glossary

Word	Meaning
Average	One summary made by adding values and dividing by how many there are.
Category	A group of related things or ideas.
Classifier	A model that identifies categories in material such as writing.
Climate	Weather patterns studied over many years.
Data	Recorded information used to investigate a question.
Emissions	Substances released by activities.
Estimate	A value worked out from available evidence, with limits.
GDP per person	Economic output divided by resident population.
Network	Points joined by connections.
Pattern	A relationship or feature found across examples.
Sample	A selection from a larger collection.
Synthetic	Constructed for a purpose, such as teaching an AI.

Missing differs from zero. Zero is an observed answer in a comparison; missing means unavailable. Understanding the question and source helps us read the picture.

Artist appendix / The 100 languages

The classifier is designed to classify writing in these **100 languages**, and the synthetic teaching dataset is multilingual. This roster is an artist reference, not a vocabulary list children need to learn. Read alphabetically down each column.

Languages 1-25	Languages 26-50	Languages 51-75	Languages 76-100
Afrikaans	Galician	Latvian	Sinhala
Albanian	Georgian	Lingala	Slovak
Amharic	German	Lithuanian	Slovenian
Arabic	Greek	Luxembourgish	Somali
Armenian	Guarani	Macedonian	Spanish
Assamese	Gujarati	Malagasy	Sundanese
Azerbaijani	Haitian Creole	Malay	Swahili
Basque	Hausa	Malayalam	Swedish
Belarusian	Hebrew	Maltese	Tajik
Bengali	Hindi	Marathi	Tamil
Bosnian	Hungarian	Mongolian	Telugu
Bulgarian	Icelandic	Nepali	Thai
Burmese	Igbo	Norwegian Bokmål	Tigrinya
Catalan	Indonesian	Odia	Turkish
Cebuano	Irish	Oromo	Turkmen
Chinese	Italian	Pashto	Ukrainian
Croatian	Japanese	Persian	Urdu
Czech	Javanese	Polish	Uyghur
Danish	Kannada	Portuguese	Uzbek
Dutch	Kazakh	Punjabi	Vietnamese
English	Khmer	Quechua	Welsh
Estonian	Kinyarwanda	Romanian	Xhosa
Filipino	Korean	Russian	Yiddish
Finnish	Kyrgyz	Serbian	Yoruba
French	Lao	Shona	Zulu

The roster does not establish equal reliability in every language, dialect, script, or mixed-language form. Norwegian Bokmål is the Norwegian entry; Chinese and Arabic each count as one broad entry. Source-language hints and the exhibition's chosen display languages are different lists.

References and further reading

These sources support the artist notes and shared factual foundation checked on 9 September 2026. Country context is not proof of why a particular display value occurs. The selected health and environmental release details remain explicitly unresolved.

Public datasets and model

- [S1] **Stanford OVAL CCNews**. Approximate upstream size, preparation, and dates. [Source page](#).
- [S2] **Twitter Sentiment by Country**. Country-associated social-media source. [Source page](#).
- [S3] **DEMOTEC participatory-budgeting tweets**. Topic-based collection, 2015-2021. [Source record](#).
- [S4] **Yachay Seasons**. Coordinate-linked texts and broader coverage. [Source page](#).
- [S5] **CFPB Consumer Complaint Database**. Consumer complaint collection. [Source page](#).
- [S6] **CMU GeoText**. Geotagged microblog collection released in 2010. [Source page](#).
- [S7] **IHME Global Burden of Disease**. Health-estimation source family. [Source overview](#).
- [S8] **WorldPop Global 2000-2020**. Population data family, including roughly 1 km data. [Product overview](#). [Population data](#).
- [S9] **Berkeley Earth**. Temperature data and product explanations. [Data overview](#).
- [S10] **Climate TRACE**. Emissions data and definitions. [Data downloads](#). [Inventory definitions](#).
- [S11] **World Bank GDP per capita**. Current US dollars; the exhibition uses 2024 values. [Indicator](#).
- [S12] **BAAI BGE-M3**. Multilingual encoder used by the classifier. [Model overview](#).

Country and health context

- [C1] **IPCC AR6 Working Group I, Chapter 4**. Regional warming and Arctic amplification. [Chapter](#).
- [C2] **World Bank, Burundi overview**. Economic development context. [Country overview](#).
- [C3] **WHO, Low back pain**. Condition and age patterns, 19 June 2023. [Fact sheet](#).
- [C4] **STATEC, Kaleidoscope 2006**. Cross-border workers and GDP-per-capita accounting. [Publication](#).
- [C5] **WHO, Osteoarthritis**. Condition definition, 14 July 2023. [Fact sheet](#).
- [C6] **MedlinePlus, Rheumatoid arthritis**. Immune-related inflammatory condition. [Medical encyclopedia](#).
- [C7] **WHO, Migraine and other headache disorders**. Everyday effects and public-health burden. [Fact sheet](#).
- [C8] **Statistics Bureau of Japan**. Population estimates for 1 October 2024. [Population overview](#).